Robust Angle-Uncertainty-Aware Joint ISAC Waveform Design and Simulation

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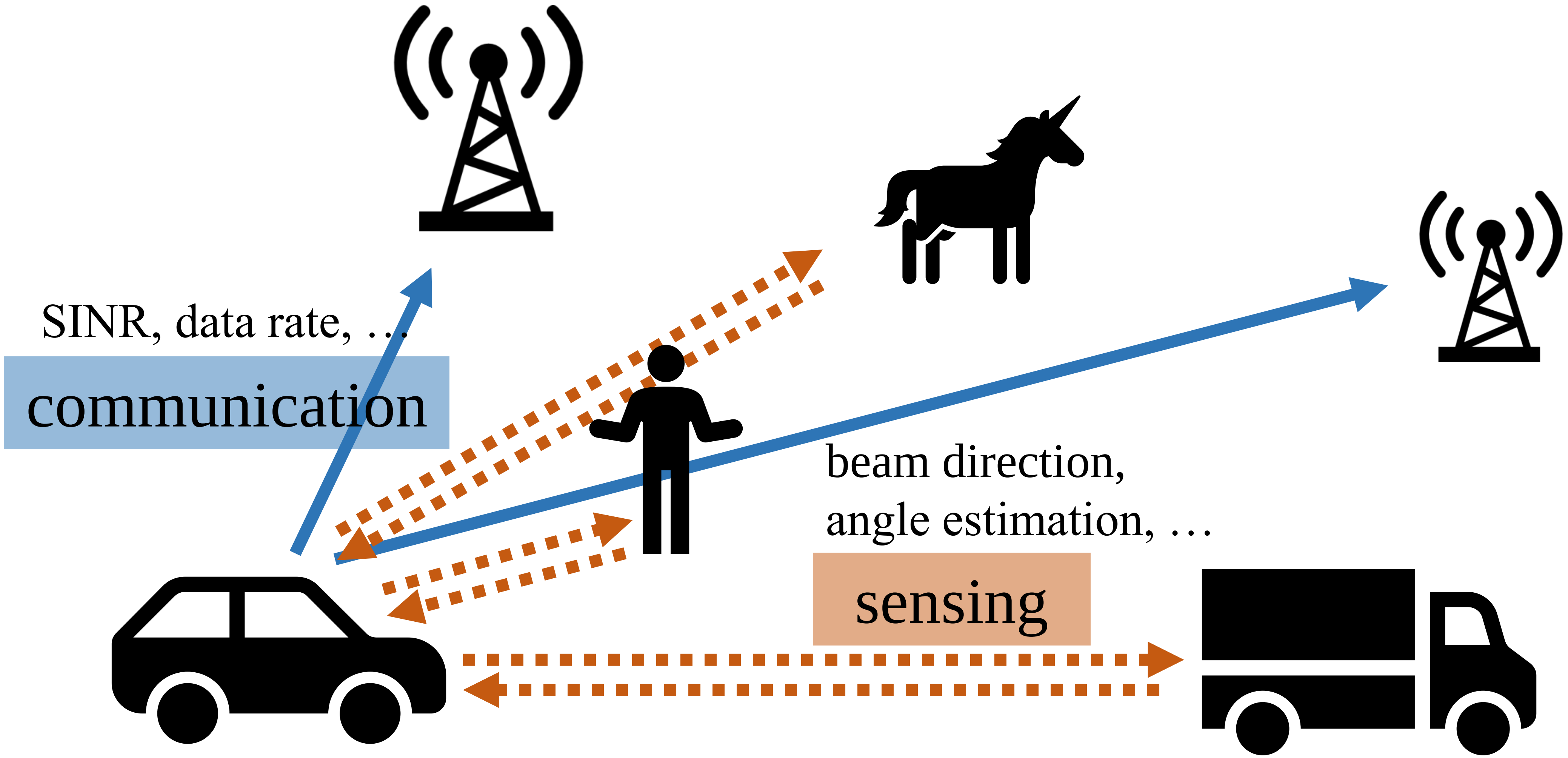
ISAC: Communication + Sensing

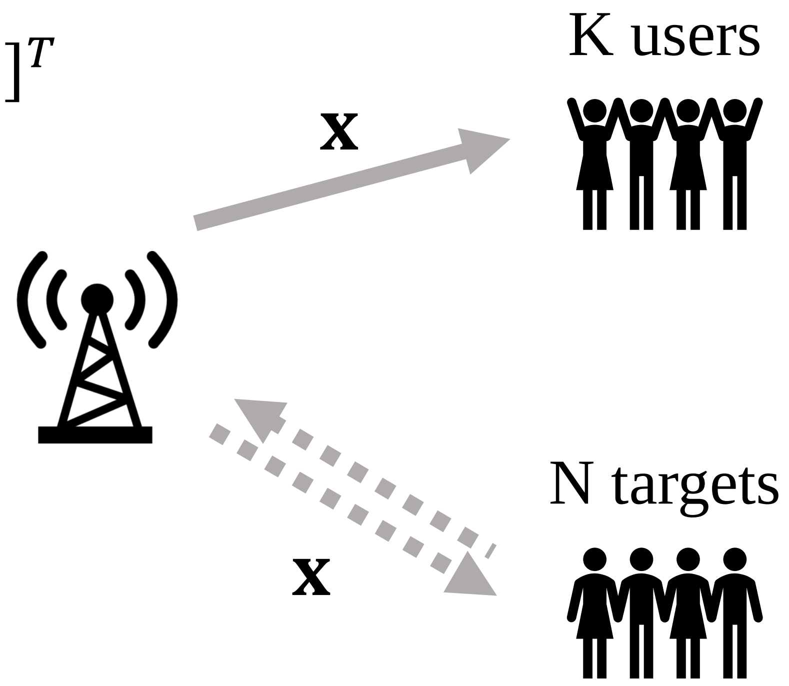


Outline

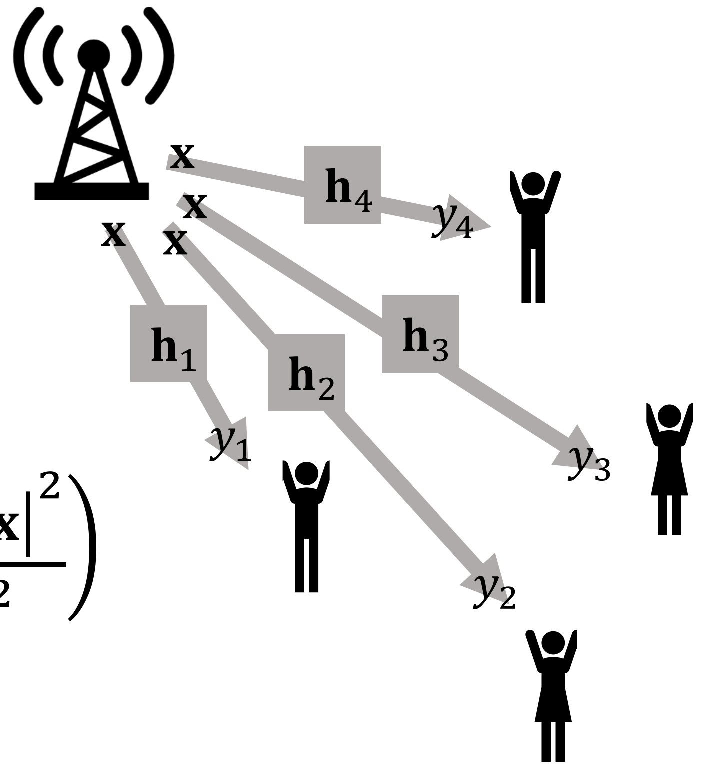
- Transmit Signal Model
- Communication Model
- Sensing Model
- Multi-Objective Problem
- Relaxation
- Simulation

Transmit Signal Model

- Transmit waveform: $\mathbf{x} = \mathbf{W}\boldsymbol{s} = \sum_{k=1}^{K} \mathbf{w}_k \, s_k$
  - Precoding matrix: $\mathbf{W} = [\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_K]$
  - Communication symbol vector: $\mathbf{s} = [s_1, s_2, \dots, s_K]^T$
- Power constraint: $\mathrm{tr}(\mathbf{R}) = \| \mathbf{x} \|_2^2 \leq P_{\max}$
  - $\mathbf{R} = \mathbf{x}\mathbf{x}^H \Rightarrow \mathbf{R} \succcurlyeq 0, \mathrm{rank}(\mathbf{R}) = 1$



Communication Model



- Received signal of user k : $y_k = \mathbf{h}_k^H \mathbf{x} + n_k$
  - Downlink channel: $\mathbf{h}_k \in \mathbb{C}^{N_t \times 1}$
  - Noise: $n_k \sim \mathcal{CN}(0, \sigma^2)$
- Achievable rate of user k: $R_k = B\log_2\left(1 + \frac{|\mathbf{h}_k^H \mathbf{x}|^2}{\sigma^2}\right)$
- Sum rate: $R_{\text{sum}}(\mathbf{x}) = \sum_{k=1}^{K} R_k$
- Communication objective in minimization form: $f_C(\mathbf{x}) = -R_{\text{sum}}(\mathbf{x})$

Constructive Interference (CI) Constraints

- Noiseless received signal: $\hat{y}_k = \mathbf{h}_k^H \mathbf{x} = \mathbf{h}_k^H \mathbf{w}_k s_k + \sum_{i \neq k} \mathbf{h}_k^H \mathbf{w}_i s_i$
- Rotated by the intended symbol:

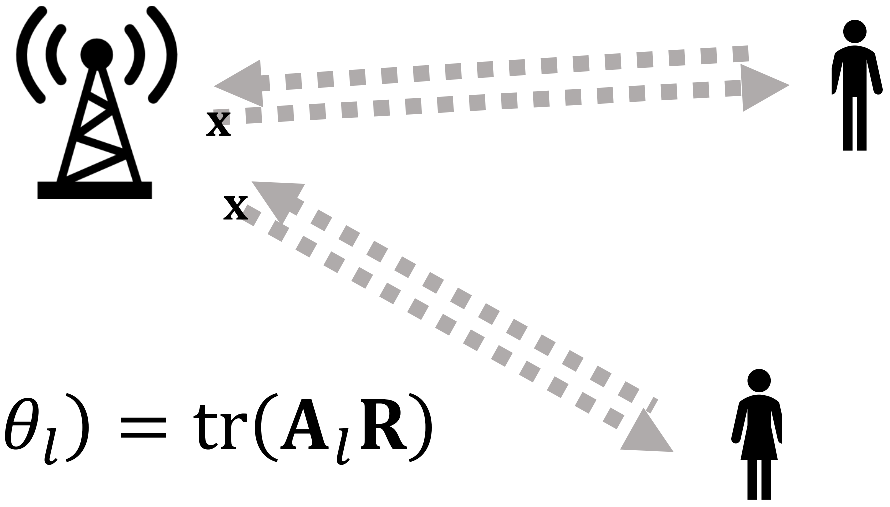
$$u_k = \hat{y}_k s_k^* = \mathbf{h}_k^H \mathbf{w}_k |s_k|^2 + \sum_{i \neq k} \mathbf{h}_k^H \mathbf{w}_i s_i s_k^*$$

- CI region with SNR threshold Γ_k:

$$\mathrm{Re}(u_k) \geq \sqrt{\Gamma_k \sigma^2},$$

$$|\mathrm{Im}(u_k)| \leq \left(\mathrm{Re}(u_k) - \sqrt{\Gamma_k \sigma^2}\right) \tan\phi, \qquad \phi = \frac{\pi}{M}$$

Sensing Model



- Transmit beampattern at grid point θ_l:

$$G(\theta_l) = |\mathbf{a}_t^H(\theta_l)\mathbf{x}|^2 = \mathbf{a}_t^H(\theta_l)\mathbf{R}\mathbf{a}_t(\theta_l) = \mathrm{tr}(\mathbf{A}_l\mathbf{R})$$

  - Transmit steering vectors: $\mathbf{a}_t(\theta)$
  - Define: $\mathbf{A}_l = \mathbf{a}_t(\theta_l)\mathbf{a}_t^H(\theta_l)$

- Desired beampattern at grid point θ_l with an error $\boldsymbol{\delta}$:

$$d_l(\boldsymbol{\delta}) = \begin{cases} 1, & \exists n \in \mathcal{N} \text{ such that } \left|\theta_l - \left(\hat{\theta}_n + \delta_n\right)\right| \leq \Delta_\theta \\ 0, & \text{otherwise} \end{cases}$$

  - Estimated target angle: $\hat{\theta}_n$. True target angle: $\hat{\theta}_n + \delta_n$
  - Half beam width: Δ_θ

Ellipsoidal Angle Uncertainty

- True target angle: $\theta_n = \hat{\theta}_n + \delta_n$
- Target angle error: $\boldsymbol{\delta} = [\delta_1, \delta_2, \dots, \delta_N]^T$

- Ellipsoidal uncertainty set: $\mathcal{U} = \{\boldsymbol{\delta} \mid \boldsymbol{\delta} = \mathbf{P}\mathbf{u},\ \| \mathbf{u} \|_2 \leq 1\}$
- For diagonal uncertainty:

$$\mathbf{P} = \mathrm{diag}(\epsilon_1, \epsilon_2, \dots, \epsilon_N)$$

$$\Rightarrow \boldsymbol{\delta} = \begin{bmatrix} \epsilon_1 u_1 \\ \epsilon_2 u_2 \\ \vdots \\ \epsilon_N u_N \end{bmatrix}, \qquad \theta_n(\mathbf{u}) = \hat{\theta}_n + \epsilon_n u_n, \qquad \| \mathbf{u} \|_2 \leq 1$$

Sensing Objective with Worst-Case MSE

- Beampattern matching MSE for any $\boldsymbol{\delta} \in \mathcal{U}$:

$$M_s(\alpha_s, \mathbf{R}; \boldsymbol{\delta}) = \frac{1}{L}\sum_{l=1}^{L} |\operatorname{tr}(\mathbf{A}_l \mathbf{R}) - \alpha_s d_l(\boldsymbol{\delta})|^2$$

  - Compensation for magnitude mismatch: α_s

- Sensing objective in minimization form:

$$f_S(\alpha_s, \mathbf{R}) = \max_{\boldsymbol{\delta} \in \mathcal{U}} M_s(\alpha_s, \mathbf{R}; \boldsymbol{\delta}) = \zeta$$

  - Worst-case consideration
  - Epigraph variable ζ: $M_s(\alpha_s, \mathbf{R}; \boldsymbol{\delta}) \leq \zeta,\ \forall \boldsymbol{\delta} \in \mathcal{U}$

Multi-Objective Problem

- Two objectives: $\min_{\mathbf{x},\alpha_s,\mathbf{R}} \begin{bmatrix} f_C(\mathbf{x}) \\ f_S(\alpha_s,\mathbf{R}) \end{bmatrix}, \begin{cases} f_C(\mathbf{x}) = -R_{\mathrm{sum}}(\mathbf{x}) \\ f_S(\alpha_s,\mathbf{R}) = \zeta \end{cases}$

- Augmented weighted Tchebycheff problem:

$$\min_{\mathbf{x},\alpha_s,\mathbf{R},\zeta}\ \max_{i\in\{1,2\}} \omega_i\left[\bar{f}_i + \xi\left(\bar{f}_C + \bar{f}_S\right)\right]$$

  - Weights: $\omega_1 \geq 0,\ \omega_2 \geq 0,\ \omega_1 + \omega_2 = 1$
  - Normalized objectives: $\bar{f}_C(\mathbf{x}) = \frac{f_C(\mathbf{x}) - f_C^*}{|f_C^*| + \varepsilon_0},\ \bar{f}_S(\zeta) = \frac{\zeta - f_S^*}{|f_S^*| + \varepsilon_0}$
  - Epigraph variable α_T: $\omega_1\left[\bar{f}_C + \xi\left(\bar{f}_C + \bar{f}_S\right)\right] \leq \alpha_T,\ \omega_2\left[\bar{f}_S + \xi\left(\bar{f}_C + \bar{f}_S\right)\right] \leq \alpha_T$
  - Small augmentation parameter: $\xi > 0$

Final Optimization Problem

$$\min_{\mathbf{x},\mathbf{R},\alpha_s,\zeta,\alpha_T} \ \alpha_T$$

$$\text{s.t.}$$

$$\omega_1\left[\bar{f}_C(\mathbf{x})+\xi\left(\bar{f}_C(\mathbf{x})+\bar{f}_S(\zeta)\right)\right]\le\alpha_T,\qquad \omega_2\left[\bar{f}_S(\zeta)+\xi\left(\bar{f}_C(\mathbf{x})+\bar{f}_S(\zeta)\right)\right]\le\alpha_T$$

$$\mathrm{Re}\left(\mathbf{h}_k^H\mathbf{x}s_k^*\right)\ge\sqrt{\Gamma_k\sigma^2},\qquad \begin{cases}\mathrm{Im}\left(\mathbf{h}_k^H\mathbf{x}s_k^*\right)\le\left[\mathrm{Re}\left(\mathbf{h}_k^H\mathbf{x}s_k^*\right)-\sqrt{\Gamma_k\sigma^2}\right]\tan\phi\\ -\mathrm{Im}\left(\mathbf{h}_k^H\mathbf{x}s_k^*\right)\le\left[\mathrm{Re}\left(\mathbf{h}_k^H\mathbf{x}s_k^*\right)-\sqrt{\Gamma_k\sigma^2}\right]\tan\phi\end{cases}$$

$$\frac{1}{L}\sum_{l=1}^{L}\left|\mathrm{tr}(\mathbf{A}_l\mathbf{R})-\alpha_s d_l(\boldsymbol{\delta})\right|^2\le\zeta,\qquad\forall\boldsymbol{\delta}\in\mathcal{U}$$

$$\mathrm{tr}(\mathbf{R})\le P_{\max},\qquad \mathbf{R}=\mathbf{x}\mathbf{x}^H,\qquad \mathbf{R}\succcurlyeq 0$$

SDR and Waveform Recovery

- Relaxation: $\mathbf{R} = \mathbf{x}\mathbf{x}^H \Rightarrow \mathbf{R} \succcurlyeq \mathbf{x}\mathbf{x}^H$
- By the Schur complement: $\begin{bmatrix} \mathbf{R} & \mathbf{x} \\ \mathbf{x}^H & 1 \end{bmatrix} \succcurlyeq 0$

- SDR tightness is checked by: $\rho_{rank} = \frac{\lambda_1(\mathbf{R}^*)}{\sum_i \lambda_i(\mathbf{R}^*)}$, $\rho_{res} = \frac{\|\mathbf{R}^* - \mathbf{x}^* \mathbf{x}^{*H}\|_F}{\|\mathbf{R}^*\|_F}$
  - $\rho_{rank} \approx 1$ and $\rho_{res} \approx 0$: $\mathbf{x}^*$ is used
  - Otherwise: Gaussian randomization generates candidate waveforms from $\tilde{\mathbf{x}} \sim \mathcal{CN}(\mathbf{0}, \mathbf{R}^*)$

Need for Simplified Simulation

Not convex in **x**!!!

$$f_C(\mathbf{x}) = -R_{\text{sum}}(\mathbf{x}) = -\sum_{k=1}^{K} B\log_2\left(1 + \frac{\left|\mathbf{h}_k^H\mathbf{x}\right|^2}{\sigma^2}\right)$$

$$\min_{\mathbf{x},\mathbf{R},\alpha_S,\zeta,\alpha_T} \alpha_T$$

$$\text{s.t.}$$

$$\omega_1\left[\bar{f}_C(\mathbf{x}) + \xi\left(\bar{f}_C(\mathbf{x}) + \bar{f}_S(\zeta)\right)\right] \le \alpha_T, \qquad \omega_2\left[\bar{f}_S(\zeta) + \xi\left(\bar{f}_C(\mathbf{x}) + \bar{f}_S(\zeta)\right)\right] \le \alpha_T$$

$$\text{Re}\left(\mathbf{h}_k^H\mathbf{x}s_k^*\right) \ge \sqrt{\Gamma_k\sigma^2}, \qquad \begin{cases} \text{Im}\left(\mathbf{h}_k^H\mathbf{x}s_k^*\right) \le \left[\text{Re}\left(\mathbf{h}_k^H\mathbf{x}s_k^*\right) - \sqrt{\Gamma_k\sigma^2}\right]\tan\phi \\ -\text{Im}\left(\mathbf{h}_k^H\mathbf{x}s_k^*\right) \le \left[\text{Re}\left(\mathbf{h}_k^H\mathbf{x}s_k^*\right) - \sqrt{\Gamma_k\sigma^2}\right]\tan\phi \end{cases}$$

$$\frac{1}{L}\sum_{l=1}^{L}\left|\text{tr}(\mathbf{A}_l\mathbf{R}) - \alpha_s d_l(\boldsymbol{\delta})\right|^2 \le \zeta, \qquad \forall\boldsymbol{\delta} \in \mathcal{U}$$

$$\text{tr}(\mathbf{R}) \le P_{\max}, \qquad \begin{bmatrix} \mathbf{R} & \mathbf{x} \\ \mathbf{x}^H & 1 \end{bmatrix} \succcurlyeq 0, \qquad \mathbf{R} \succcurlyeq 0$$

Simplified Simulation Problem

Not convex in $\mathbf{x}$!!!

$$
\begin{aligned}
f_C(\mathbf{x}) &= -R_{\text{sum}}(\mathbf{x}) \\
&= -\sum_{k=1}^{K} B\log_2\left(1 + \frac{\left|\mathbf{h}_k^H\mathbf{x}\right|^2}{\sigma^2}\right)
\end{aligned}
$$

$$
\min_{\mathbf{x},\mathbf{R},\alpha_s,\zeta,\alpha_T} \quad \alpha_T
$$

$$
\text{s.t.}
$$

$$
\cancel{\omega_1\left[\bar{f}_C(\mathbf{x}) + \zeta\left(\bar{f}_C(\mathbf{x}) + \bar{f}_S(\zeta)\right)\right] \le \alpha_T,} \qquad \cancel{\omega_2\left[\bar{f}_S(\zeta) + \zeta\left(\bar{f}_C(\mathbf{x}) + \bar{f}_S(\zeta)\right)\right] \le \alpha_T}
$$

$$
\mathrm{Re}\left(\mathbf{h}_k^H\mathbf{x}s_k^*\right) \ge \sqrt{\Gamma_k\sigma^2}, \qquad
\begin{cases}
\mathrm{Im}\left(\mathbf{h}_k^H\mathbf{x}s_k^*\right) \le \left[\mathrm{Re}\left(\mathbf{h}_k^H\mathbf{x}s_k^*\right) - \sqrt{\Gamma_k\sigma^2}\right]\tan\phi \\
-\mathrm{Im}\left(\mathbf{h}_k^H\mathbf{x}s_k^*\right) \le \left[\mathrm{Re}\left(\mathbf{h}_k^H\mathbf{x}s_k^*\right) - \sqrt{\Gamma_k\sigma^2}\right]\tan\phi
\end{cases}
$$

$$
\frac{1}{L}\sum_{l=1}^{L}\left|\mathrm{tr}(\mathbf{A}_l\mathbf{R}) - \alpha_s d_l(\boldsymbol{\delta})\right|^2 \le \zeta, \qquad \forall\boldsymbol{\delta} \in \mathcal{U}
$$

$$
\mathrm{tr}(\mathbf{R}) \le P_{\max}, \qquad \begin{bmatrix} \mathbf{R} & \mathbf{x} \\ \mathbf{x}^H & 1 \end{bmatrix} \succcurlyeq 0, \qquad \mathbf{R} \succcurlyeq 0
$$

Simplified Simulation Problem

$$\min_{\mathbf{x},\mathbf{R},\alpha_{\mathrm{s}},\zeta} \ \zeta$$

$$\mathrm{s.t.}$$

$$\mathrm{Re}\left(\mathbf{h}_k^H \mathbf{x} s_k^*\right) \geq \sqrt{\Gamma_k \sigma^2}, \qquad \begin{cases} \mathrm{Im}\left(\mathbf{h}_k^H \mathbf{x} s_k^*\right) \leq \left[\mathrm{Re}\left(\mathbf{h}_k^H \mathbf{x} s_k^*\right) - \sqrt{\Gamma_k \sigma^2}\right] \tan\phi \\ -\mathrm{Im}\left(\mathbf{h}_k^H \mathbf{x} s_k^*\right) \leq \left[\mathrm{Re}\left(\mathbf{h}_k^H \mathbf{x} s_k^*\right) - \sqrt{\Gamma_k \sigma^2}\right] \tan\phi \end{cases}$$

$$\frac{1}{L}\sum_{l=1}^{L} |\operatorname{tr}(\mathbf{A}_l \mathbf{R}) - \alpha_{\mathrm{s}} d_l(\boldsymbol{\delta})|^2 \leq \zeta, \qquad \forall \boldsymbol{\delta} \in \mathcal{U}$$

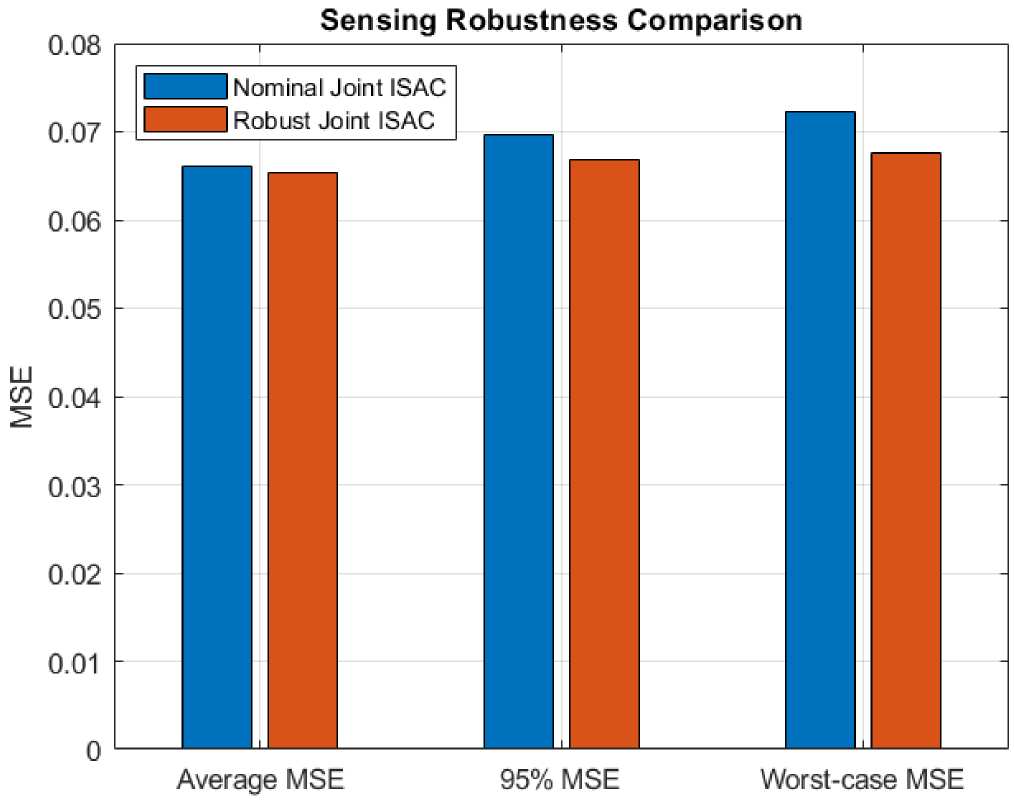
$$\operatorname{tr}(\mathbf{R}) \leq P_{\max}, \qquad \begin{bmatrix} \mathbf{R} & \mathbf{x} \\ \mathbf{x}^H & 1 \end{bmatrix} \succcurlyeq 0, \qquad \mathbf{R} \succcurlyeq 0$$

Robust vs. Nominal Joint ISAC

- 2 targets
- $\mathbf{P} = \mathrm{diag}(5, 3)$

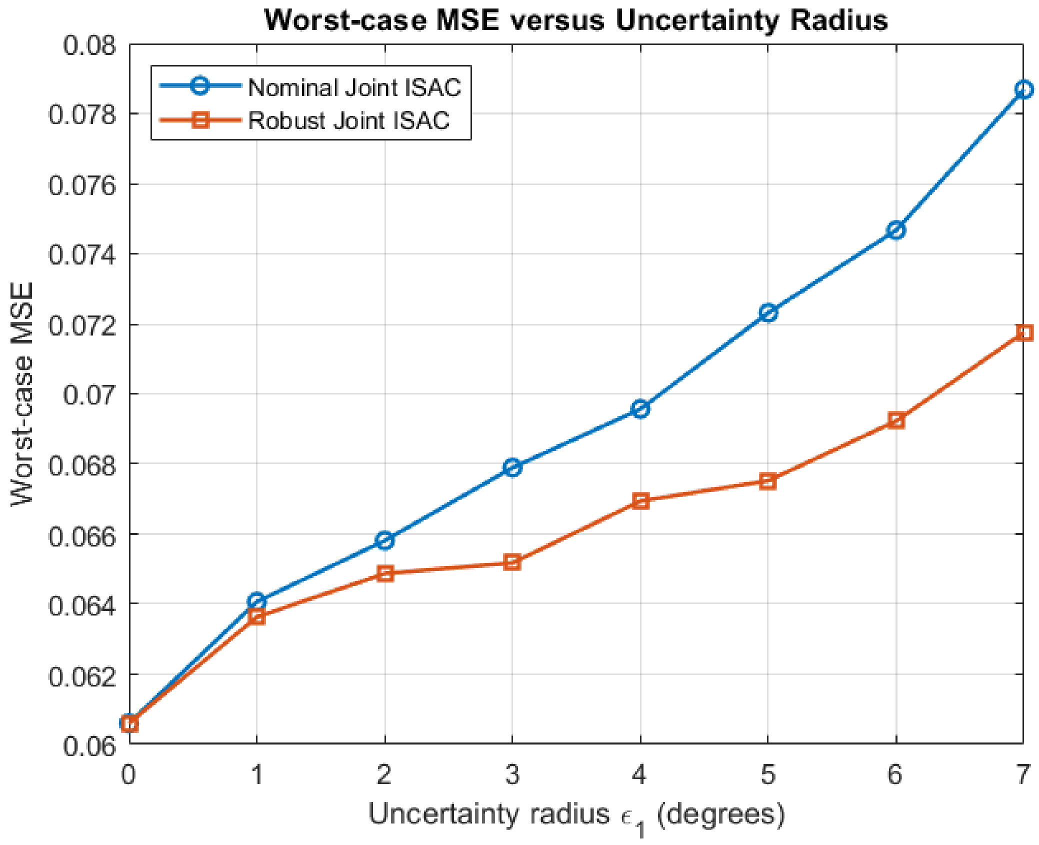
Method	SumRate	NominalMSE	AvgMSE	P95MSE	WorstMSE	CI
Nominal Joint ISAC	7.6456	0.060602	0.066157	0.069569	0.072315	true
Robust Joint ISAC	7.5223	0.062395	0.065399	0.066878	0.067525	true

Method	ρ_{rank}	ρ_{res}	Recovery
Nominal Joint ISAC	1.0000	8.7217×10^{-9}	Direct optimized $\mathbf{x}$
Robust Joint ISAC	1.0000	3.6727×10^{-9}	Direct optimized $\mathbf{x}$



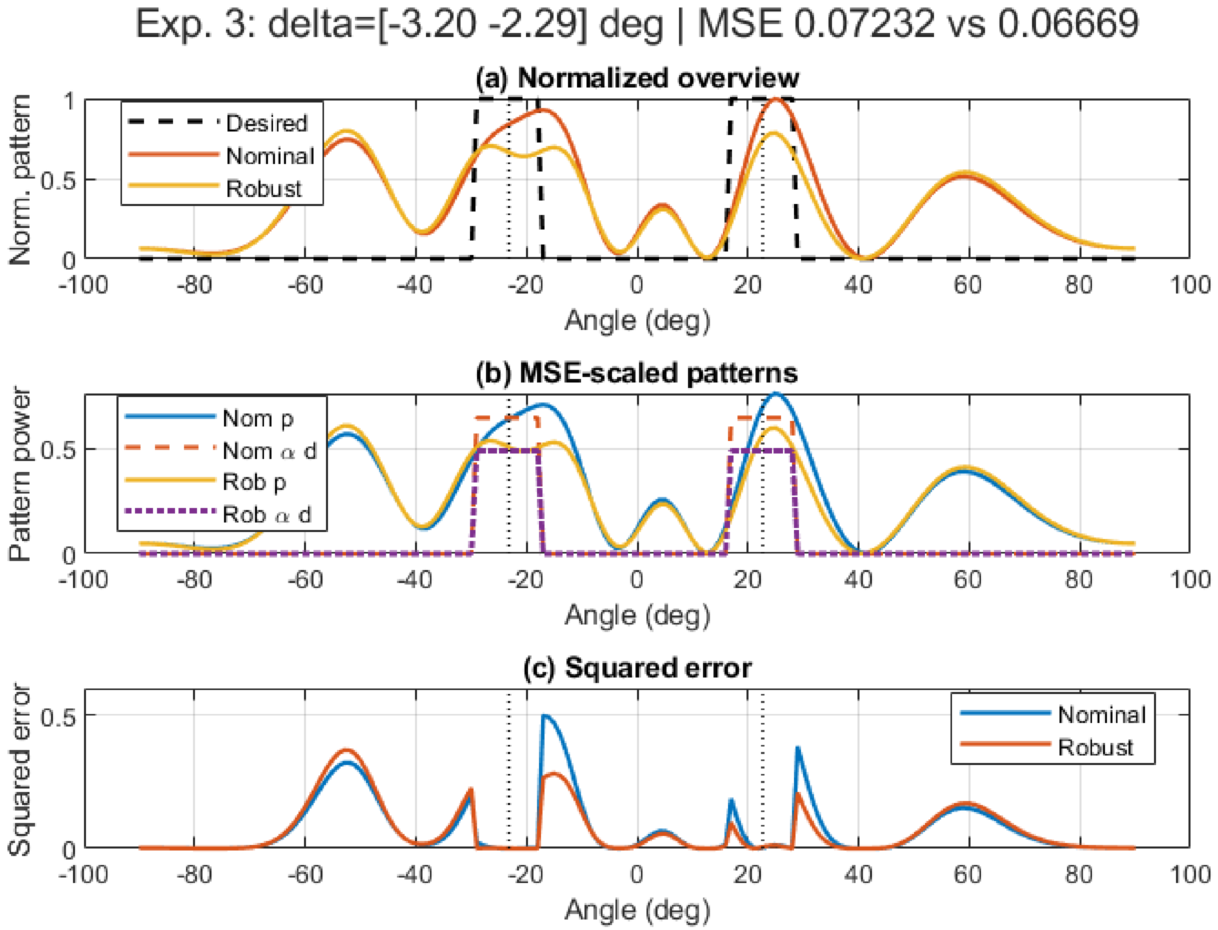
Worst-Case MSE versus Uncertainty Radius

- 2 targets
- $\mathbf{P} = \mathrm{diag}(\epsilon_1, 0.6\epsilon_1)$



Beampattern Under Worst-Case Angle Error

- Worst-case sample from Experiment 1
- 2 targets
- Estimated angle: $-20°, 25°$
- True angle: $-23.2°, 22.71°$



Thank you for listening

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